

CAN AI LEARN TO SPOT MALIGNANCY?

Breast-mass FNA classification | Pink AI 2026

Relax Team

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Our model is the least interesting part of our work.
The data had a trap in almost every column.



THE DATA WAS LYING

01

A column
that leaks
the answer

02

Columns that
contradict
geometry

03

Patients
entered twice,
with opposite
diagnoses

04

A test set
drawn
differently
from train

TRAP 1

The leak

The column `biopsy_followup_code` nearly reproduces the diagnosis, so we dropped it.



96.1%

ONC-REF: malignant

2.1%

ROUTINE: malignant

Alone it scores

AUC 0.971

Written after the diagnosis,
so a model would learn the clerk, not the cells

Decision: dropped

TRAP 2

A GEOMETRY LIE DETECTOR

A nucleus obeys geometry:

$$P \approx 2\pi r \text{ and } A \approx \pi r^2$$

CONSTANTS
LEARNED
FROM DATA

6.479

3.080

158 rows: perimeter and area swapped

1378 cells: radius in cm instead of mm

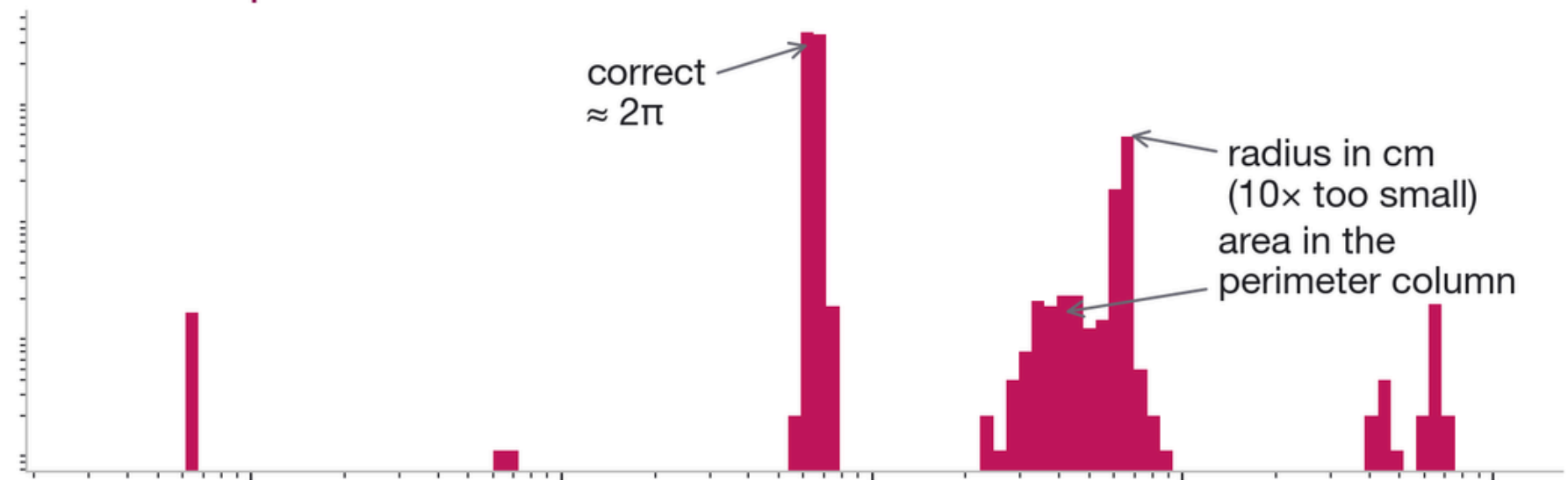
40 negative areas, wrong even after abs()

2601 missing size cells rebuilt from geometry

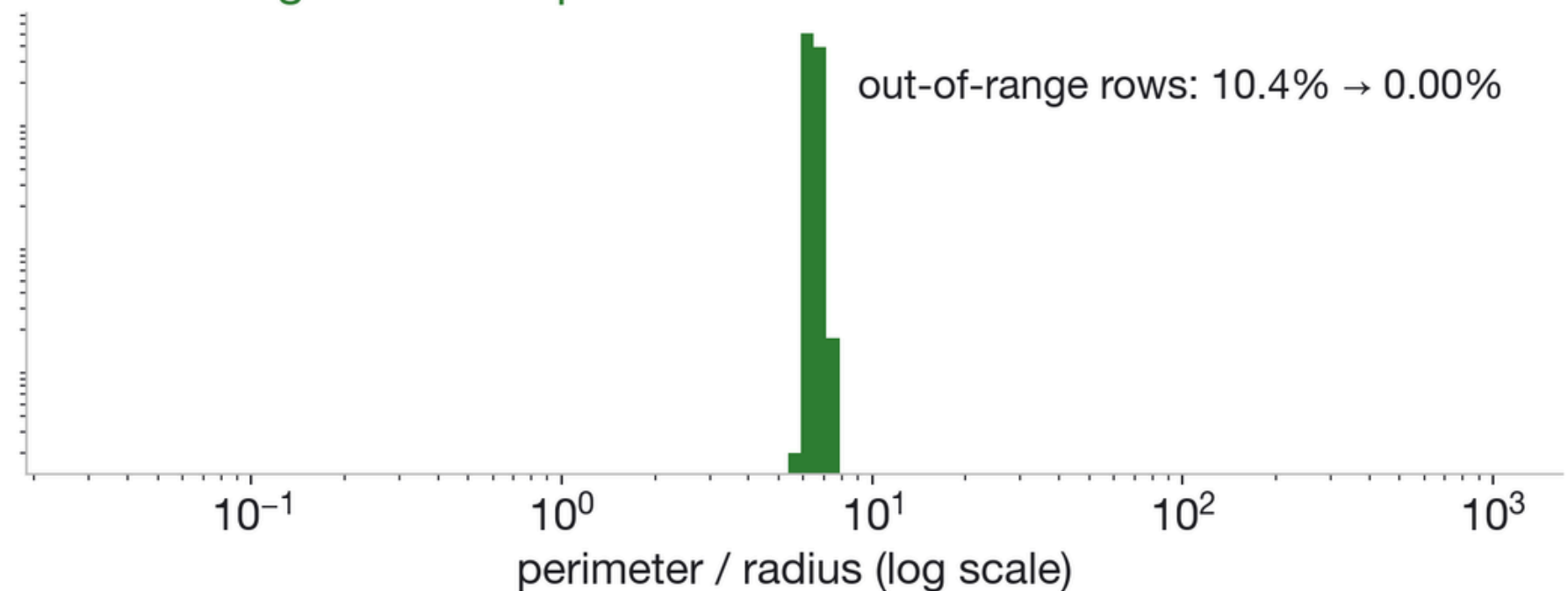
No rows deleted

Impossible P/R ratios: 10.4% → 0.00%

BEFORE: perimeter / radius



AFTER the geometric repair



TRAP 3

DUPLICATES AND NOISY LABELS

346

patients
entered
twice

50

with opposite
diagnoses

48

labels
repaired

Merged; 48 labels
repaired with the
follow-up code

97.7%

agreement on
uncontested rows

About 3-6% of training
labels look randomly
flipped

MODEL AND VALIDATION

ENSEMBLE MODEL	Logistic Regression + Gradient Boosting ensemble
LEAK-PROOF CV	StratifiedGroupKFold 5 × 3, grouped by patient no patient appears in both train and validation
RESULT	CV AUC 0.9329 ± 0.0004 scored against noisy labels
IMPACT OF CLEANING	recall at 90% precision 0.5473 → 0.7613

CV AUC

0.9329

against noisy labels

UNDER THE HOOD

Our technical stack and pipeline, end to end in Python

01 CLEAN

- pandas + regex parsing
- Decimal commas, thousands separators, ? / -- / unknown to NaN
- IDs, labels, centres and dates normalised
- Mojibake notes repaired

02 REPAIR

- Sister-column regressions fix $\times 100$ / $\times 1000$ slips
- Geometry: $P = 6.479 R$ and $A = 3.080 R^2$
- Duplicates merged, labels adjudicated
- All parameters learned on train only

03 MODEL

- scikit-learn
- LogisticRegression (L2) + HistGradientBoosting, averaged
- 30 log10 measurements + age
- StratifiedGroupKFold 5 $\times$ 3, grouped by patient

04 DECIDE

- IsotonicRegression calibration
- Label-noise correction
- Prevalence stress test at 20%
- $t = 0.84$, bootstrap-checked

TECH STACK

Python 3 · pandas · NumPy · scikit-learn · Matplotlib · Jupyter on DataCamp DataLab

No deep learning · no external labels · test set never used for fitting · notebook runs top to bottom

THE THRESHOLD

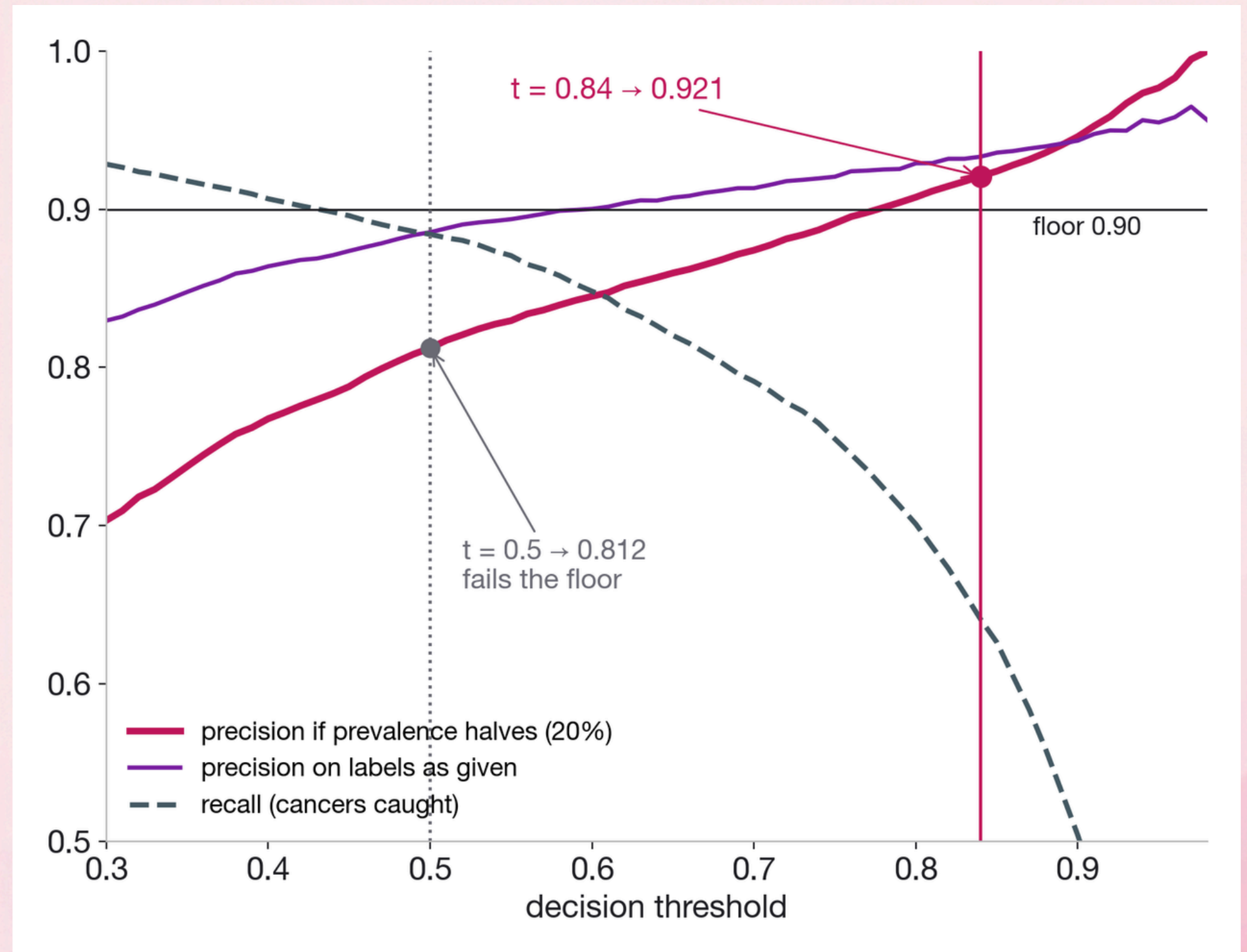
Chosen under a stress test because the test set is drawn differently.

$t=0.84$

Stress test: label noise removed
and prevalence halved to 20%

Expected precision 0.921,
recall 0.641

Default $t = 0.5$ gives precision 0.812, which
fails the 0.90 floor



LIMITS AND RESPONSIBLE USE

Weaker for women under 40
(CV AUC 0.888)

A new site or scanner needs
re-calibration (Site C reads larger)

Never a stand-alone diagnosis.
It is a triage aid for pathologists.



THANK YOU, QUESTIONS?